Math 242, S13 **Name:** Answer

Quiz 9 **SSN:** xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find a particular solution y_p of the given nonhomogeneous differential equation:

$$y'' - y' - 2y = 2x^2+3$$

Solution: The complementary solution of the above equation is

$$y_c(x) = C_1 e^{2x} + C_2 e^{-x}$$

(characteristic equation: $r^2 - r - 2 = 0$, $(r-2)(r+1) = 0$, $r_1 = 2,\ r_2 = -1$)

The function $f(x) = 3x^2+4$

and its derivatives $f'(x) = 6x$

$f''(x) = 6$

Thus none of them appears in y_c. Then we guess

$$y_p(x) = x^s(Ax^2+Bx+C) = Ax^2+Bx+C \quad (\text{Since } s=0)$$

$$\Rightarrow y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

Substitution in the original equation gives us

$$y''-y'-2y = (2A)-(2Ax+B)-2(Ax^2+Bx+C)$$

$$= (-2A)x^2 + (-2A-2B)x + (2A-B-2C) = 3x^2+4$$

$$\Rightarrow \left.\begin{array}{l} -2A = 2 \\ -2A-2B = 0 \\ 2A-B-2C = 3 \end{array}\right\} \Rightarrow \begin{array}{l} A = -1 \\ B = 1 \\ C = -3 \end{array}$$

thus $\underline{y_p(x) = -x^2+x-3}$.